

# Enhancing existing insider threat detection tools

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**Abstract**— *Insider threat is the most difficult security challenge; it comes from users who are already inside the organization and have high privileges and permission to access the system, not like external threats, which are easier to detect. Human error, carelessness are the reasons why these risks arise accidentally; they can also be caused on purpose through actions such as data theft or sabotage. The primary issue is that security measures are fully aimed at detecting external threats. As a result, organizations face risks of data leakage, financial losses, and reputational damage from data breaches. In our research, the main goal is to improve one of the existing tools of detection of insider threats that were identified in the gap analysis, by enhancing its detection accuracy and reducing false alarms. Development will focus on the optimization of data preprocessing and applying advanced models to detect abnormal behavior, expecting a more reliable detection system that helps the organization monitor insider activity, leading to high protection of the system and preventing unauthorized access.*

**Keywords**— *insider threat, Data preprocessing, security tools enhancement.*

## I. INTRODUCTION

Insider threats are considered one of the most challenging cybersecurity problems since attackers are legitimate users, making malicious action very difficult to detect [1]. Recent studies have shown that internal incidents have increased in companies and environments that rely on remote access, cloud systems, and distributed data, especially in cases where there is privilege abuse, identity misuse, and data leakage [2]. Firewalls and intrusion detection systems are not effective with authenticated users, which allows insiders to operate without raising immediate suspicion [3]. For this reason, modern detection systems have shifted toward intelligence behavior analysis and machine-learning-based monitoring. The façade model, for example, uses deep contextual anomaly detection to build user-behavior embeddings and significantly reduce false positives in real environments [1]. Also, another study shows an AI-driven insider risk management system that

used adaptive scoring and large-language-model-based threat detection to analyze user intent and detect misuse in real time [4]. Graph-based approaches have been shown to be effective in mapping relationships between users, files, and devices to identify abnormal access patterns [2]. Furthermore, systematic reviews shows that deep-learning architectures can detect subtle changes in user behavior that traditional system fail to capture [5]. Boosting the detection accuracy and lowering the rate of false alarms by deep learning, adaptive scoring, and graph-based analysis, which will also enhance the ability to respond to insider attacks before significant damage happens.

## II. Literature Review

### A. High-precision insider threat detection using deep contextual anomaly detection

The Facade system implies the deep learning technique for insider-threat detection that evaluates each user move in the light of contextual information (like position and previous behavior), thus allowing a much clearer identification of irregular activity. A major advantage of this being that it does not demand any labeled attack data—rather, it employs contrastive and contextual learning to discern anomalies from the norm—and it has the capacity to produce results with very few false positives and very high accuracy. On the downside, the method is highly reliant on comprehensive, enterprise-grade logging and is therefore not likely to be effective in smaller setups with restricted data, leading to difficulties in deployment as well as reproducibility

### B. Insider threat detection for specific threat scenarios

The current paper introduces a scenario-based insider-threat detection model that detects misuse of privileges, impersonation, and data transfer by means of multi-head

attention applied to enriched user-behavior sequences. It is accurate beyond generic methods and long-term behavioral dependencies in detection are its main advantages. On the other hand, the model's limitations are its dependence on well-labeled datasets and a drop in performance with increasing normal-to-abnormal ratios which restricts applicability in real life situations.

#### *C. Structurally Prune Anything: Any Architecture, Any Framework, Any Time*

In this article, a fresh structured pruning framework known as SPA is introduced, which is indifferent to architecture, indifferent to framework, and allows for pruning during any phase of training. The main feature of this framework is the unified pipeline: models are changed from their original form to a common computational graph using ONNX, interconnected channels are divided into groups, importance of groups is evaluated, and pruning is done accordingly—thereby making it possible to “prune any architecture, any framework, any time”. Another strength is the “OBSPA” variant which claims to achieve state of the art results without the need for fine tuning or calibration data. One of the main drawbacks is that the method, although largely applicable in theory, has been mainly validated on image classification benchmarks (CIFAR10/100 for instance) and may not completely prove the approach across different domains or very large models; furthermore, the dependency of the method on appropriate channel coupling/grouping through the computational graph might be a problem in the case of very complex or non standard architectures.

#### *D. Genome Sequence of the Plant-Growth-Promoting Endophyte *Curtobacterium flaccumfaciens* Strain W004*

In the study by Chebotar et al. (2023), the entire genome of a strain of *Curtobacterium flaccumfaciens* (W004), a winter wheat seed endophyte, was described, and its probable usefulness in plant growth was pointed out. The study used Oxford Nanopore MinION sequencing to generate a circular genome of 3.63 Mbp with 70.89% GC content and discovered biosynthetic clusters for antibiotics and siderophores which could be among the factors that lead to plant growth and stress tolerance. The experiment proved that W004 could increase the biomass of wheat by 31.4% and the yield by 54.3%, thus it could be used as a biofertilizer. The strong points of the research are: the application of long-read sequencing for the full genome assembly, exhaustive annotation of the gene clusters, and the checking of the growth-promotion effects with

experiments. The drawbacks are: a limited functional verification of the predicted biosynthetic genes, the possibility of sequencing errors due to reliance on Nanopore data, and the lack of large field trials across different plant species or environmental conditions.

#### *E. Intelligent Video Streaming at Network Edge: An Attention-Based Multiagent Reinforcement Learning Solution*

In the past, edge video streaming research has treated the allocation of edge users and the adaptive bitrate streaming of video as two independent tasks. One task decides which server will be connected by the users while the other one adjusts the quality of the video. Although these approaches increase the load balance or bitrate, respectively, they are quite ineffective in dynamic environments due to the interdependency of the two decisions—their main drawback.

The paper under review proposes the use of a multi-agent reinforcement learning model along with an attention mechanism where both tasks are handled simultaneously as a means of allowing the servers to cooperate and enhancing the overall Quality of Experience with less streaming interruption and less server activity—a significant advantage. Nevertheless, the method entails complicated training and continues to rely on simulation before real-world deployment, which is a downside.

#### *F. Evaluation and Response*

The ITDSTS model, developed by Tian et al. (2025), is a scenario-based insider threat detection system that is intended to detect privilege misuse, identity theft, and data infiltration. It employs a combination of user activity logs and text mining through TF-IDF, Isolation Forest, and a Transformer encoder with multi-head attention, thus, the model recognizes both semantic and temporal connections in user activities. This strategy boosts detection accuracy and robustness to the point of surpassing traditional approaches like LSTM and GCN which are the CERT 4.2 dataset losing their ground. Notably, though, the research depends on a synthetic dataset which affects its practical applicability and has also restricted the consideration of only three threat scenarios, thus making it less flexible in terms of new or hybrid attacks. Moreover, the Transformer framework does enhance precision, however, it also increases the computing power required along with non-interpretability in practical deployment.

### *G. Response / Mitigation*

The National Counterintelligence and Security Center has published the new 2024 NITTF Insider Threat Guide which updates the old standards to assist the U.S. agencies in developing potentially performing insider threat programs. The guide provides a systematic outline under the categories of leadership, training, data access, monitoring, etc., laying great stress on areas of privacy, legal compliance, and interagency cooperation. The major force of the guide is its wide-ranging and flexible framework which gives the agencies the opportunity to make the programs fit their respective missions. On the other hand, it is more policy-oriented than data-driven and presents very few real-world evaluations or measurable outcomes. Besides, the guide's limited relevance outside the government context has reduced its overall applicability, though still it is a major national reference for finding the right balance between security and civil rights.

### *H. The Processes of Insider Threat Analysis*

The paper “The Processes of Insider Threat Analysis” (2023) dives into the analytical and psychological processes that take place to identify insider threats in organizations. It takes a process-oriented approach, looking at how the analysts interpret behavioral indicators, contextual data, and organizational cues to identify the risk of insider threats. The most important aspect of this work is that it is human-centered, highlighting the cognitive factors, decision-making biases, and collaboration between the analysts—these are the areas that are usually ignored in the technical models. It gives a very good example of how the knowledge and context of an analyst affect the accuracy of threat detection. But the research is mostly theoretical and does not have any empirical validation or quantitative performance measures. Its limited focus on the analyst's behavior also makes it less applicable to automated or large-scale detection systems. On the other hand, it bears the human dimension of security analysis by drawing the behavioral science and analytical methodology into insider threat research and thereby emphasizing the human factor behind the security analysis.

### *I. Insider Threat Detection with Deep Neural Network*

Yuan et al. (2018) present a model for the detection of insider threats based on deep neural networks that consists of the integration of Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) to automatically identify the anomalous behaviors of users. The authors treat user actions as sequences comparable to words in a language model and, thus, the system learns temporal dependencies and abstracts behavioral patterns without manual feature engineering. The model when

tested on the insider threat dataset CERT v4.2 managed to achieve a remarkable AUC of 0.9449 which was better than the AUCs of the conventional machine learning techniques like logistic regression. The major advantage of the study is its feature learning automation, which diminishes human bias and heightens detection accuracy. Moreover, it successfully interfaces short- and long-term user behavior through its hybrid LSTM-CNN architecture. The reliance on synthetic datasets is, however, a limitation that impacts the practical implementation of the approach in the real world. Then, there is the problem of resource-intensive computation involved in deep learning models and lack of transparency over detection decisions which also continue to present barriers to practical deployment. But still, the work lays down a solid base for the use of deep learning in insider threat detection and indicates a significant move towards adaptive behavior-based security systems.

### *J. A SYSTEMATIC REVIEW ON MACHINE LEARNING INSIDER THREAT DETECTION MODELS, DATASETS AND EVALUATION METRICS*

This review is the culmination of a decade of research (2013-2023) on the different models of machine learning (ML) and deep learning (DL) for insider-threat detection, it reviews algorithms, datasets, and evaluation metrics in 22 selected studies. The main advantage of this review is its systematic, structured methodology with criteria for clear inclusion/exclusion, a well-defined search process, and comprehensive mapping showing which ML/DL methods (e.g. SVM, LR, k-NN, LSTM) and datasets (especially the CERT r4.2 dataset) are most often used. Besides, it points out common evaluation metrics (accuracy, precision, recall, AUC-ROC) and marks the big hurdles such as data imbalance and the situation where accuracy is the main metric despite its drawbacks in the case of imbalanced datasets. The drawbacks are that it depends solely on published studies (hence, it is prone to publication bias), the analysis is based on a very small number of datasets (most studies deal with CERT datasets as opposed to real organizational data), and the review shows “what is being done” rather than giving reasons for the different models' performance (good or bad) in practice insider-threats settings. Furthermore, the review notes but cannot assess the applicability of many DL methods in this area due to the lack of rich behavioral data.

### *K. Enhanced Insider Threat Detection Through Machine Learning Approach With Imbalanced Data Resolution*

Prasad et al. (2024) introduce a machine-learning based framework for the detection of insider threats that is fully

aware of the problem of class imbalance in datasets, which is normally encountered in this area of study. They implement both oversampling (e.g., SMOTE, ADASYN) and undersampling (e.g., Cluster Centroids, Edited Nearest Neighbors) techniques on a malicious insider threat dataset that is available to the public (the CERT v4.2 dataset) and then carry out a comparison between five algorithms (Logistic Regression, Decision Tree, Random Forest, AdaBoost, and Naïve Bayes). The main advantage of the research is that it explicitly addresses the issue of a data imbalance, which is usually neglected in the insider-threat research area, and that it applies ensemble techniques to increase accuracy—600% accuracy is cited with stacking/voting classifiers. Nonetheless, the method is constrained by the fact that it is based on only one synthetic dataset, and the extent to which it can be applied to actual or diverse environments remains questionable; in addition, while the performance metrics are impressive, the study pays less attention to the interpretability of the models, and the sampling methods (particularly oversampling) may lead to overfitting in underrepresented classes. In summary, the authors' work represents a significant progress in epidemic of the detection of insider threats, by blending the strategies of addressing imbalances with machine learning, however, the future research should be on non-restricted real-world datasets and on model transparency issues.

#### *L. A Theory-Based Deep Learning Approach for Insider Threat Detection*

Deep-learning insider-threat detection and classification is a method that Wanyonyi, Wafula, and Onyango (2024) propose. They use a Gated Recurrent Unit (GRU) architecture to model user network behavior and also base their method on sociological theories—essentially, Social Bond Theory and Situational Crime Prevention Theory. One key aspect of this model is that it is designed to break large volumes of data down into smaller components—characterized by dataset imbalance and high dimensionality—through preprocessing steps. Moreover, it produces satisfactory results during testing on the synthetic CERT v4.2 dataset in terms of accuracy, recall, precision, and The F-measure. Grounding the research on theory is one of the advantages, as it helps easily connect user behavior modeling to socio-psychological insight and, moreover, the use of deep learning to GRU's capturing of temporal and spatial relationships is also a plus. Another plus is the research's acknowledgment of and approach to the practical data issues of imbalance and dimensionality. In comparison, reliance on a synthetic dataset is a shortcoming that hinders the generalization to the real world, whereas deep learning models, despite their power, still face challenges of being interpretable and deployable.

Additionally, the theoretical backing might be advantageous but the research only connects socio-psychological concepts to operational detection pipelines, where they are less acknowledged, to the extent of their importance.

#### *M. Insider Threat Detection Using Behavioural Analysis through Machine Learning and Deep Learning Techniques*

Analytical methods based on machine learning (ML) and deep learning (DL) are applied to user behavior analysis for detecting insider threats as shown in recent research. For instance, Song et al. (2024) used Bidirectional LSTM networks to accurately model time-based patterns, but their method posed a computational demand and was consequently impractical for real-time deployment. Al-Shehari et al. (2024) significantly improved data balance through DBLOF but applied extensive parameter tuning as a trade-off. Graph-based methods (Nikiforova et al., 2024; Roy et al., 2024) worked well in capturing the relationships among users but required large annotated datasets.

Li et al. (2023) and Sridevi et al. (2024) achieved impressive results with hybrid ML–DL methods that incorporated both traditional and deep learning models, however, they could not generalize the results beyond a particular dataset. LSTM and GRU (Villarreal et al., 2023; Meng et al., 2021) methods were characterized by strong anomaly detection but also experienced high false-positive rates and class imbalance. Human-centered models (Whitelaw et al., 2024) provided psychological and linguistic features but the main challenge was nearly no scalability.

#### *N. Insider Threat Detection Using Machine Learning Approach*

Insider-threat detection has started making more use of machine learning and deep learning, which are the current trend in technology, as rule-based systems were not able to catch insiders who were behaving like normal users. Significant accuracy was reported by prior studies using Random Forest, SVM, LSTM, and CNN models, indicating the power of ML/DL in the area of behavioral patterns recognition. Noteworthy is that a majority of investigations heavily depend on manual feature extraction that is time-consuming and hard to adjust when log structures change, and deep models also have difficulties when it comes to pruned datasets containing very few malicious actions. The present research is a step ahead of the above-mentioned limitations as it applies automated feature extraction (DFS), dimensionality reduction (PCA), and SMOTE for

dataset balancing thus cutting down on human effort but also raising the performance of the detection system.

#### *O. Classifier Suites for Insider Threat Detection*

Developed by Tuor and collaborators in the year 2019, the suggested technique is an unsupervised deep neural network framework that uses log data from the CERT dataset to detect insider threats. The model was very powerful as it combined recurrent neural networks (RNNs) with autoencoders to capture the normal usage patterns of users and identify abnormal ones without needing any labeled data. The whole approach gained attention because of its proficiency to cope with very intricate temporal relations and along with that it was also able to minimize the need for manually produced features thus acquiring one of the most notable strengths of traditional rule-based systems.

Upsides of the highlighted work among others include dialing up to large enterprise logs, making an effective use of unsupervised learning for the very rare insider events, and the advantage of being able to cope with changing user patterns. At the same time, the downside is its limited interpretability owing to deep models that usually function as black boxes, and its susceptibility to noise or benign anomalies in the data. To add more, since it depends on synthetic datasets like CERT, the question of real-world generalization remains unanswered.

#### *P. Machine learning based Insider Threat Modelling and Detection*

Le and Zincir-Heywood (2019) introduced a new supervised machine learning framework for the detection of insider threats by using the CERT r5.2 dataset. The entire process of their method was to check user behavior on different time scales (session, day, and week) and to apply models like logistic regression, random forest, and neural networks for spotting wicked activity. Among the varied features of this study, the most significant ones were its multi-granularity analysis which enhanced the detection of threats thus providing more flexibility, and its user-based evaluation that gave more practical results of the detection. Moreover, it conducted a test on the model generalization across different versions of the dataset, thus, giving more strength to the findings.

On the downside, the restrictions of the work included a heavy class imbalance, lower accuracy on new data, and low explainability of the complex models used. Moreover, the use of the synthetic CERT data set limited the applicability of the research in real-world scenarios. The study, therefore, pointed out the need for even better

handling of class imbalance and more transparency in the models but at the same time contributed significantly in terms of associating behavioral patterns with detection accuracy.

#### *Q. Insider Threat Detection using Deep Autoencoder and Variational Autoencoder Neural Networks*

In their work, Pantelidis et al. (2019) came up with an autoencoder (AE) and variational autoencoder (VAE) based method for detecting insider threats using the CERT r4.2 dataset. The deep learning models used by them have automatically learned the behavioral representations without having to go through extensive feature engineering and have also been able to attain very high accuracy in the detection of malicious activities. Detection of complicated behavioral anomalies, high accuracy in the detection, and very little dependence on the manual feature design are some of the advantages of this study. On the other hand, the study encountered some limitations such as lack of interpretability, reliance on synthetic data, and possible issues of generalization in the case of application to real-world data. In conclusion, the study pointed out that deep unsupervised models for insider threat detection are promising and at the same time, it raised the issue of requiring more transparent and adaptable frameworks.

#### *R. Anomaly Detection for Insider Threats Using Unsupervised Ensembles*

Le & Zincir-Heywood (2021) present an innovative and highly promising unsupervised anomaly detection technique for insider threats, which is based on a strong combination of four different unsupervised models (autoencoder, isolation forest, LODA, LOF), several temporal data representations (concatenation, percentile difference, mean/median difference), and ensemble strategies (averaging, max, voting). The real-world scenario of having unlabelled insider-behaviour data is tackled through the application of the unsupervised methods, which is a very strong point of the paper. The second point is, the temporal representations of user activity are explored to capture changes in behaviour over time. The results are very strong indeed; for instance, aberrant users are detected at a budget of 0.1% with a success rate of ~60%, and all malicious ones under 5% of the budget across several datasets (including CERT R4.2, CERT R6.2, LANL), which also shows ensemble superiority. The weak points consist of (i) still depending on synthetic or semi-synthetic benchmark datasets (like CERT) for validations instead of large volumes of truly real-world organizational data, which limits generalizability, (ii) the method is unsupervised but still

relies on training assumptions (that normal behavior prevails, anomalies are rare) which may not be the case in the dynamic insider-threat environments, and (iii) though temporal representations help to detect, the chosen representations (windowing, concatenation) may not be capable of uncovering the complex latent motives or context of user behavior changes, so the detection of new or sophisticated insider threat tactics is restrained.

#### *S. Problem Definition and Data gathering*

In their study, Smith and his colleagues (2013) presented the CMU CERT Insider Threat Dataset, a publicly available dataset which is aimed at aiding research into insider threats as the main purpose of the dataset. The dataset is composed of various kinds of logs like user logons, file access, emails, and device usage combined with simulated scenarios like data exfiltration, sabotage, and credential theft. Its most important advantage is the possibility of creating a reproducible benchmark through which researchers can test and compare detection methods across the whole range of user behaviors. The dataset's unique features such as different log types and realistic threat scenarios gain it momentum in being utilized in a number of studies that come afterward. One of the negatives of this dataset is that the insider activities are artificially created rather than being drawn from actual incidents, making it less realistic. It also lacks the organizational background details and it depicts outdated IT environments which could consequently impact the generalizability of the findings to present day real-world systems. Notwithstanding these drawbacks, the dataset continues to be an essential instrument for the progress of insider threat detection research.

#### *T. Data Preprocessing*

Haidar et al. (2018) unveiled AnyThreat, a system for insider threat detection that put more emphasis on the detection of one anomalous behaviour of the insider rather than all the anomalous behaviours. The method makes use of role-based communities, an oversampling technique (AMOTRE) to cope with class imbalance and class decomposition to enhance classification. The main advantage is the ability to pragmatically reduce false positives while allowing high detection rates, which is a key to solving the difficulties in insider threat detection. Nevertheless, the technique used relies heavily on synthetic datasets and feature engineering which may compromise the realism of the study and its applicability to the real world. Moreover, the opportunistic strategy that is being used, although it has been effective in minimizing false

alarms, it might miss the quiet insiders whose actions are not easy to notice or are hidden. Even with these drawbacks, AnyThreat is still a new and practical detection method that can be applied in such situations where there is a high need for the accuracy of detection with low false positives.

#### *U. Model Design*

Ali et al. (2025) introduced a real-time detection framework for insider threats that integrates behavioral analytics with deep evidential clustering. The system takes user activity logs of a certain period and transforms them into latent embeddings, then it uses very certain clustering to spot unusual behavior. It is capable of accurately modeling uncertainty, hence it has the great feature of reducing false positives, it also adjusts to the changes in human behavior and very accurately detects unnoticeable insider threats. But the study conducted that evaluated the system was based on existing benchmark data and not on real deployments which could be a reason for the generalizability issue. Another problem is that the model is complicated, involving a combination of the steps of temporal encoding, clustering, and drift detection, which may again pose challenges of implementation and tuning in the practical world. Yet, this research is a significant step forward in the area of uncertainty-aware, behavior-based insider threat detection, and it lays a good ground for further research and operational systems.

#### *V. Adaptive Thresholding*

This research paper ranks among the most detailed review articles in the area of anomaly detection and it presents a systematic classification of statistical, distance-based, and machine-learning techniques. The writers stress that the traditional static-threshold systems are of no use in changing environments as they are unable to change with kindred user behavior or data patterns. The research has drawn attention to the need for adaptive thresholding wherein the thresholds would be set according to the real-time behavioral pattern as opposed to a set of predefined rules. This is indicative of the fact that dynamic, data-driven thresholds are going to greatly reduce the number of false alarms and thus enhanced the accuracy of anomaly detection.[21]

#### *W. Contextual Feature Extraction*

This citation highlights the critical necessity of adding contextual features to the existing methods of network

anomaly detection. The writers claim that the use of basic metrics only (like packet counts or traffic volume) will not be enough to detect the advanced anomalies. They suggest examining the contextual attributes like time of the day, frequency, access sensitivity and privilege levels that let the models understand normal operational behavior much better. The review gives powerful proof that the contextual feature extraction improves both precision and reliability in abnormal user actions detection.[22]

#### *X. Machine Learning-Based Dynamic Scoring*

This paper presents a discussion about the impact of data mining and machine learning on the evolution of cybersecurity systems, which are no longer static but have become adaptive and intelligent. The authors emphasize the capability of algorithms such as Random Forest, Naïve Bayes, and Support Vector Machines (SVM) to not only learn from past scenarios but also to change the risk evaluations in real time. The authors' conclusions lead to the introduction of dynamic scoring models instead of the conventional threshold-based systems whereby user actions are rated and grouped into risk categories according to learned behavioral patterns. This, in turn, highlights the importance of automatic machine-learning risk scoring as a method for detecting insider threats.[23]

#### *Y. Explainability of Alerts*

The paper that is provided highly influences and it also delivers LIME (Local Interpretable Model-Agnostic Explanations), which is a method for making the decision of the machine learning classifiers understandable. The authors mention the difficulty of black-box models in highly important areas like cybersecurity, where the need for unambiguous justification for alert signals is very high. LIME makes it interpretable and transparent by showing the model's prediction depending on the features that contributed the most. This indirectly leads to the development of explicable alerts, giving security analysts insights into the reasons behind the user's action being flagged and thus increasing trust and speeding up incident response.[24]

#### *Z. Risk-Level Output (Low/Medium/High)*

The presented study evaluates the methodologies of detecting anomalies in network security and discusses the drawbacks of the binary alert systems that categorize events merely as "attack" or "normal." The researchers claim that the risk scoring of incidents based on levels such as low, medium, and high provides much greater value in real-

world working environments. The advantages of such a classification are that security personnel can rank their reactions, distribution of resources can be done properly, and alert fatigue can be evaded. The study is very much in favor of the enhancement of anomaly detection systems with the risk-level outputs in order to make the right decision faster and improve the overall response efficiency.[25]

### III. GAP ANALYSIS

A comparative analysis of sixteen typical studies on insider threat detection is presented in Table I. The chosen works represent a rather broad methodological range, such as policy and governance structures, statistical and machine-learning (ML) models, deep-learning (DL) architectures, and anomaly-based detection systems. This comparison aims at finding some variation in methodology, performance, datasets and implementation focus and finding some common limitations in current approaches. The current analysis can be used to develop an all-encompassing framework that would include both organizational and technical insider threat mitigation measures.

| REF  | Detection Method                                                                                       | Accuracy            | Performance                                                                   | Dataset                                         | Weaknesses                                                                                                                | Strengths                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| [2]  | TF-IDF + Transformer Encoder + Isolation Forest for Scenario-based Insider Threat Detection            | 95 %<br>(F1 = 0.94) | Excellent memory and accuracy; tailored for situation-specific categorization | CERT Insider Threat Dataset                     | 1. Limited validation in the actual world<br>2. It might not apply to all companies.<br>3. Excessive computational demand | 1. Superior detection for certain situations<br>2. Accuracy and recall in balance<br>3. Combines behavioral and semantic analysis               |
| [6]  | Insider Threat Governance Based on Policy and Programs (NITTF Framework)                               | N/A                 | N/A (framework qualitative)                                                   | None (policy document)                          | 1. The absence of a technological model<br>2. No empirical support<br>3. No emphasis on datasets                          | 1. A thorough guide to mitigation and governance<br>2. Outlines duties and functions<br>3. Harmonizes organizational and technological controls |
| [7]  | Insider Threat Analysis Analytical Process Model (SEI Framework)                                       | N/A                 | Data → Indicators → Risk Evaluation is the conceptual process flow.           | None (conceptual model)                         | 1. Absence of quantifiable measurements<br>2. The absence of ML integration<br>3. Difficult to automate                   | 1. A solid conceptual basis<br>2. Directs the integration of technological models<br>3. Encourages policy coherence                             |
| [8]  | CNN and LSTM in a Deep Neural Network for User Behavior Sequences                                      | AUC = 0.9449        | Effective sequence modeling for temporal and spatial feature learning         | CERT Insider Threat Dataset V4.2                | 1. Potential overfitting 2. Not in real time<br>3. Just pay attention to logs                                             | 1. Recognizes sequential dependencies<br>2. Excellent precision<br>3. Integration of deep learning                                              |
| [9]  | Behavioral Features for the Identification of Anomalies (KTH Study)                                    | ~90 %<br>(F1≈0.89)  | Anomaly-scoring methodology; moderate runtime                                 | Custom Insider Log Dataset                      | 1. A high percentage of false positives<br>2. Threshold sensitivity<br>3. Inadequate behavioral variance                  | 1. Identifies unidentified insider activity<br>2. Lightweight<br>3. Adaptable and unsupervised                                                  |
| [10] | Logistic Regression, Random Forest, AdaBoost, and Naïve Bayes ensemble machine learning using sampling | 99.5 %              | properly manages unbalanced data                                              | CERT Insider Threat Dataset                     | 1. Potential overfitting 2. Dependency on synthetic data<br>3. Is not explicable                                          | 1. Deals with the disparity in class<br>2. Evaluates algorithms<br>3. Extremely precise                                                         |
| [11] | Deep Learning (MobileNetV2, VGG19, ResNet50 Transfer Learning)                                         | ≈97 %               | Robust image-feature extraction at a high computational cost                  | Behavior-Image Dataset (transformed audit logs) | 1. Computationally demanding<br>2. Poor interpretability<br>3. Requires GPU power                                         | 1. Superb generalization<br>2. Robust feature representation<br>3. Innovative approach based on images                                          |
| [12] | ML and DL (Organizational Logs) in Behavioral Analysis                                                 | ≈94 %               | integrates DL and ML for behavioral analytics                                 | User Log Dataset (unspecified)                  | 1. An unclear dataset<br>2. Partial measurements<br>3. Poor repeatability                                                 | 1. A emphasis on behavior<br>2. Integrates techniques<br>3. Useful and flexible design                                                          |
| [13] | Insider Threat Detection Using ML Classifiers (Bin Sarhan & Altwaijry 2023)                            | ~94 %               | Accuracy that is consistent across tested algorithms                          | CERT Dataset variant                            | 1. Data on a small scale<br>2. Does not use real-time testing<br>3. The possibility of overfitting                        | 1. Multiple machine learning comparisons<br>2. Easy execution<br>3. A solid foundation investigation                                            |
| [14] | Insider Threat Detection Classifier Suites (Noever 2019)                                               | —                   | 88 algorithms are compared broadly (best ≈ 95%).                              | CERT Insider Threat Dataset                     | 1. Surface-level analysis for each model<br>2. No validation of deployment<br>3. Pay attention to static benchmarks       | 1. Extensive benchmarking<br>2. Determines the top-performing algorithms<br>3. Analysis of feature importance                                   |
| [15] | ML-Based Insider Threat Modeling and Detection (Multilevel approach)                                   | ~97 %               | Activity profiling using a multilevel data model                              | CERT Dataset                                    | 1. Inadequate validation across domains<br>2. Concentrate on a particular dataset<br>3. Only static actions               | 1. Adaptable structure<br>2. High rate of detection<br>3. Comparing different approaches                                                        |
| [16] | Variational and Deep Autoencoder Networks (Pure Portsmouth 2021)                                       | 96 %<br>(F1≈0.93)   | Robust categorization based on reconstruction errors                          | Internal Log Dataset (Portsmouth)               | 1. Expensive training<br>2. Contextual purpose may be missed<br>3. The black-box model                                    | 1. Feature learning without supervision<br>2. Identifies abnormalities without labeling<br>3. Manages data that is not linear                   |
| [17] | An overview of current developments in the detection of malicious insider threats (AIP 2022)           | —                   | ML/DL trends and performance ranges are identified by a literature study.     | Multi-source survey                             | 1. Absence of empirical research<br>2. Only summarized stats<br>3. Non-experimental                                       | 1. A thorough synopsis<br>2. Emphasizes difficulties<br>3. Identifies potential lines of inquiry                                                |
| [18] | Documentation for the SEI Insider Threat Dataset (CMU Library)                                         | —                   | The process of creating and describing datasets                               | CERT Insider Threat Dataset (all versions)      | 1. Artificial actions<br>2. Insufficient realism<br>3. Pay attention to the corporate domain                              | 1. A common benchmark<br>2. Extensive metadata<br>3. Facilitates model assessment                                                               |
| [19] | Preprocessing and Feature Engineering of Insider Threat Data (arXiv 1812.00257)                        | —                   | explains the CERT dataset's preparation.                                      | CERT Dataset (CMU)                              | 1. No assessment of the model<br>2. Just concentrate on data cleansing<br>3. Inadequate metrics for accuracy              | 1. Important for reproducibility<br>2. Specifies the data workflow<br>3. Makes model training easier                                            |
| [20] | Deep Evidential Clustering for Insider Threat Identification in Real Time (arXiv 2505.15383)           | 94.7 %              | reduces false positives by around 38%.                                        | CERT and TWOS datasets                          | 1. A complicated system<br>2. Requires practical testing<br>3. Resource-intensive                                         | 1. Adaptation in real time<br>2. Manages ambiguity<br>3. Cuts down on false alarms                                                              |

Table 1: Gap analysis

#### A. Policy- and Governance-Based Approaches.

The studies conducted by the National Insider Threat Task Force (NITTF) and the Software Engineering Institute (SEI) focus on organizational governance and formal insider threat management systems. These works characterize the basic policies, reporting, and data-handling rules that are needed in order to facilitate detection efforts. Though all these methods cannot be measured in terms of detection metrics, they offer the necessary advice on how to roll out sustainable insider threat programs. Their significant weakness is that they are not integrated with automated analyzing systems, but they create the procedural and ethical background of the technical research.

#### B. Detection Approaches that are based on machine learning.

Research like Bin Sarhan and Altwaijry (2023), Noever (2019), and others recommends the use of algorithms such as Random Forest, AdaBoost, Logistic Regression, and Naive Bayes as the basis of developing insider detection systems in the framework of the implementation of machine learning. The accuracy levels are usually between 94 and 99 percent, proving the fact that ML models can identify behavioral anomalies in an efficient manner. Although these are encouraging findings, the majority of the models make extensive use of CERT, which is not a real organizational setting, but a simulation. These systems are therefore highly accurate in controlled studies but with fewer generalizability in the operational environment. Their interpretability and small cost of computation are still crucial benefits to their use in business settings.

#### C. Detection Approaches Done by Deep-Learning.

The recent literature shows that there is a switch towards the use of DL models, such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Autoencoders, and transfer-learning models, like ResNet50 and VGG19. These models are good at learning temporal and spatial correlations among activity logs and system events of users with reported AUC and accuracy of more than 0.94. They are however complex and this poses challenges in explainability and computational efficiency. Nevertheless, DL methods have been found to outperform traditional ML in detecting insider behaviors that are subtle and evolving, as well as exhibit a great deal of flexibility to large-scale data.

#### D. Detection Approaches that are based on Anomalies and Behaviors.

The Kth (2025) work and other studies take advantage of unsupervised learning to detect abnormal user behavior by an anomaly-based detection model. These methods do not have many labelled data needs, which makes them suitable in tracking bad habits that are unknown. However, their performance is also determined by threshold choice and heterogeneity of behavior and this can lead to high false-

positive rates. Their flexibility and scalability, however, can be useful in dynamism of enterprise settings.

#### E. Real-Time Adaptive Models and Hybrid E. Hybrid and Real-Time Adaptive Models.

The use of hybrid frameworks (ML, DL, and behavioral analytics) has recently been presented as an effective solution to the shortcomings of single-method frameworks. Methods that combine Transformer encoders, Isolation Forest, or Evidential Clustering show that they both achieve higher precision and lower false-positive rates, as well as can adapt in real-time. These systems focus on contextual awareness, which combines technical indicators and behavioural indicators to offer a more holistic defence mechanism against insider threats. Their main weakness is in computation complexity and the need to have on-going data streams.

### IV. PROPOSED FRAMEWORK

In this section, the improved framework that was created to overcome the shortcomings identified in prior research. While [9] established a deep learning foundation. Our framework enhances it by addressing its static thresholds and lack of contextual awareness. The enhancement solutions will minimize the false positives [14], maximize accuracy, and improve the interpretability of alerts.

#### A. Behavioral Scoring Tool Origin Limitation.

The initial model applied in the previous studies transforms the behavior of the user into a score of anomaly and compares it with a predetermined value. When the score goes above this value, the behavior is reported as suspicious. Yet, the method has a number of limitations: Established thresholds result in high false-positive rates. Absence of contextual awareness (time, sensitivity, frequency, or history of user). Score fails to conform to behavioral changes. Low explainability: The analysts are not able to know why a given event was given a red flag. These shortcomings prompted the creation of a better scoring system.

#### B. Suggested Improved Behavioral Anomaly Scoring Framework.

The suggested framework is a paradigm shift of strict and rule-defined detection to a dynamic, smart, and open system. It is designed on five pillars that are interrelated:

**Adaptive & Dynamic Thresholding:** Instead of fixed limits, the framework dynamically uses thresholds that can be refractively updated based on a series of user actions and the context [1, 2], which reduces the number of false alarms by a significant margin [10].

**Context-Sensitive Feature Enrichment:** It takes multi-dimensional data in the form of time based patterns, sensitivity of data, and the order of actions [3, 17, 19] to create a more accurate behavioral model going beyond the simplified analysis of logs [4].

**Machine Learning-Based Behavioral Modeling:** A core ML model (e.g., Deep Neural Network) is trained on historical data to learn complex patterns of behavior [8, 9, 10], and thus can use it to compute anomaly scores, as indicative of subtle, contextual risks.

**Representable, Risk-Tiered Alerting:** Alerts are ranked (Low/Medium/High) to be effectively responded to [10, 12] and have a clear explanation (e.g., after-hours access to sensitive files) through explainable AI methods [24], which makes analysts trust them.

**Continuous Learning Feedback Loop:** The system uses the feedback of the analysts to continuously improve its models and user baselines [15, 20]. This makes it be able to change to new behaviors and emerging threats beating the brittle nature of previous systems [21].

### C. Anticipated Impact

This combined strategy aims at providing:

- Reduced False Positives through adaptive thresholds & context [10, 14].
- Increased Detection Accuracy by ML modeling [8, 9].
- More efficient use of the analysts with explained, prioritized alerts [24].
- Lasting Adaptability through unremitting learning [15, 20].
- Consistency with the current security plans of proactive defense [5, 6].

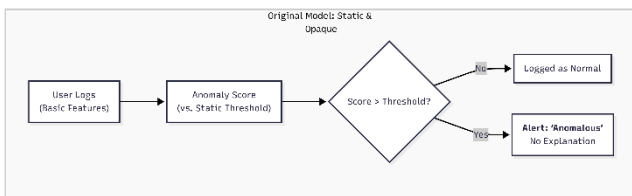


Figure 1: Flowchart of the Baseline Model (Based on Reference [9])

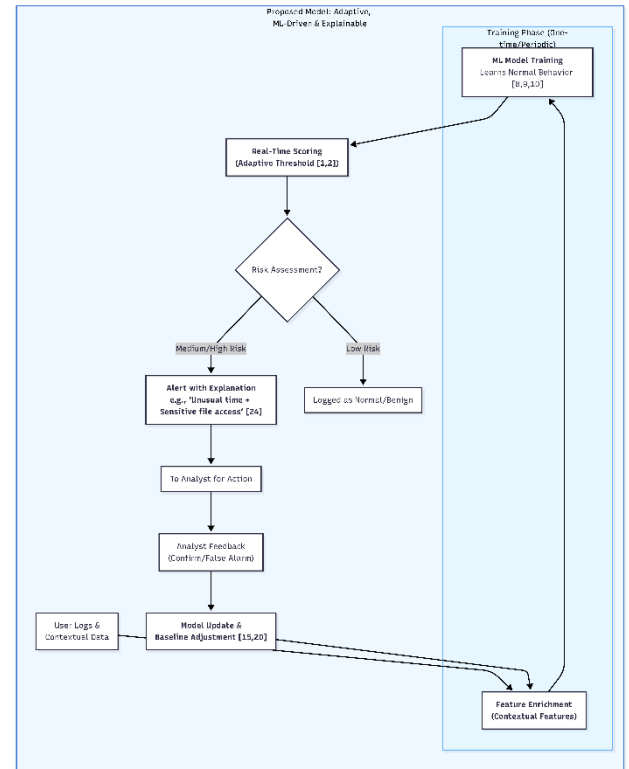


Figure 2: Flowchart of the Proposed Enhanced Framework

| feature                          | Original tool                  | Proposed enhanced tool                                              |
|----------------------------------|--------------------------------|---------------------------------------------------------------------|
| <b>Threshold Type</b>            | Static fixed threshold         | Adaptive threshold based on behavior [1,2]                          |
| <b>Features Used</b>             | Basic behavioral features only | Basic + contextual features (patterns, time, sensitivity) [3,17,19] |
| <b>False Positive Rate</b>       | High                           | Reduced via adaptive thresholds & context [1,2,10]                  |
| <b>Scoring Method</b>            | Simple anomaly score           | ML-based scoring with behavioral modeling [8,9,10]                  |
| <b>Awareness of User Context</b> | Very limited                   | High context awareness through feature enrichment [3,17]            |
| <b>Response Output</b>           | Binary alert                   | Risk-tiered alerts (Low/Med/High) [10,12]                           |
| <b>Explainability</b>            | Not provided                   | Explainable alerts with reasons [24]                                |
| <b>Generalization</b>            | Low                            | Improved via continuous learning [15,20]                            |

Table 2: Original vs. Enhanced Behavioral Anomaly Scoring Systems

## V. CONCLUSION

In summary, this research provides an enhanced model based on behavioral anomaly scoring model to mitigate insider threat. This enhanced model provides thresholding, feature extraction, and machine learning- based dynamic scoring. By addressing the limitation of original model, these enhancements aim to reduce false positives, improve alert accuracy, and provide explanations which make analyzing more easily. So, the enhancement model offers a more effective and reliable solution to detect insider threats.

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